Lab 3: Branch Prediction Using SimpleScalar

Computer Architecture Laboratory

1 Objective

This lab studies the effect of branch prediction schemes on branch address prediction rate and processor CPI/IPC. The lab uses `sim-bpred` for branch predictor statistics and `sim-outorder` for timing impact. The structure follows the UMass SimpleScalar branch prediction lab, with paths updated for the current repository layout:

```
http://www.ecs.umass.edu/ece/koren/architecture/Simplescalar/branch_prediction.htm
```

Run all commands from the cloned repository root.

```
cd /path/to/SimpleScalar
```

2 Setup

Configure for Alpha because this lab uses the Alpha benchmarks from Lab 1:

```
make clean
make config-alpha
make
mkdir -p labs
```

Get help for `sim-bpred`:

```
./sim-bpred -h
```

Important predictor option:

```
-bpred {nottaken|taken|bimod|2lev|comb}
```

Useful statistics:

Statistic pattern	Meaning
`bpred_taken.bpred_addr_rate`	Branch address prediction rate for taken predictor.
`bpred_nottaken.bpred_addr_rate`	Branch address prediction rate for not-taken predictor.
`bpred_bimod.bpred_addr_rate`	Branch address prediction rate for bimodal predictor.

Statistic pattern	Meaning
`*.bpred_dir_rate`	Branch direction prediction rate.

3 Part 1: Branch Address Prediction Rate

Run each benchmark using three prediction schemes: taken, not taken, and bimodal.

3.1 anagram.alpha

```
./sim-bpred -bpred taken -redir:sim labs/bpred_anagram_taken.out benchmarks/anagram.alpha
     benchmarks/words < benchmarks/anagram.in
./sim-bpred -bpred nottaken -redir:sim labs/bpred_anagram_nottaken.out benchmarks/anagram
    .alpha benchmarks/words < benchmarks/anagram.in
./sim-bpred -bpred bimod -redir:sim labs/bpred_anagram_bimod.out benchmarks/anagram.alpha
     benchmarks/words < benchmarks/anagram.in
```

3.2 go.alpha

```
./sim-bpred -bpred taken -redir:sim labs/bpred_go_taken.out benchmarks/go.alpha 50 9
    benchmarks/2stone9.in
./sim-bpred -bpred nottaken -redir:sim labs/bpred_go_nottaken.out benchmarks/go.alpha 50
    9 benchmarks/2stone9.in
./sim-bpred -bpred bimod -redir:sim labs/bpred_go_bimod.out benchmarks/go.alpha 50 9
    benchmarks/2stone9.in
```

3.3 compress95.alpha

```
./sim-bpred -bpred taken -redir:sim labs/bpred_compress_taken.out benchmarks/compress95.
    alpha < benchmarks/compress95.in
./sim-bpred -bpred nottaken -redir:sim labs/bpred_compress_nottaken.out benchmarks/
    compress95.alpha < benchmarks/compress95.in
./sim-bpred -bpred bimod -redir:sim labs/bpred_compress_bimod.out benchmarks/compress95.
    alpha < benchmarks/compress95.in
```

3.4 cc1.alpha

```
./sim-bpred -bpred taken -redir:sim labs/bpred_cc1_taken.out benchmarks/cc1.alpha -O
    benchmarks/1stmt.i
./sim-bpred -bpred nottaken -redir:sim labs/bpred_cc1_nottaken.out benchmarks/cc1.alpha -
    O benchmarks/1stmt.i
./sim-bpred -bpred bimod -redir:sim labs/bpred_cc1_bimod.out benchmarks/cc1.alpha -O
    benchmarks/1stmt.i
```

Extract prediction rates:

```
grep -E "bpred_.*bpred_addr_rate|bpred_.*bpred_dir_rate" labs/bpred_*.out
```

Fill Table 1.

Benchmark	Taken branch address prediction rate	Not-taken branch address prediction rate	Bimod branch address prediction rate
`anagram.alpha`			
`go.alpha`			
`compress95.alpha`			
`cc1.alpha`			

Create a histogram for the four benchmarks and draw a conclusion. Does the result agree with your intuition?

4 Part 2: CPI Impact Using `sim-outorder`

Use **`sim-outorder`** to calculate CPI for two benchmarks. The original lab skips the other two because they can take longer to simulate. Run:

```
./sim-outorder -config config/default.cfg -bpred taken -redir:sim labs/ooo_anagram_taken.
    out benchmarks/anagram.alpha benchmarks/words < benchmarks/anagram.in
./sim-outorder -config config/default.cfg -bpred nottaken -redir:sim labs/
    ooo_anagram_nottaken.out benchmarks/anagram.alpha benchmarks/words < benchmarks/
    anagram.in
./sim-outorder -config config/default.cfg -bpred bimod -redir:sim labs/ooo_anagram_bimod.
    out benchmarks/anagram.alpha benchmarks/words < benchmarks/anagram.in
```

```
./sim-outorder -config config/default.cfg -bpred taken -redir:sim labs/ooo_compress_taken
    .out benchmarks/compress95.alpha < benchmarks/compress95.in
./sim-outorder -config config/default.cfg -bpred nottaken -redir:sim labs/
    ooo_compress_nottaken.out benchmarks/compress95.alpha < benchmarks/compress95.in
./sim-outorder -config config/default.cfg -bpred bimod -redir:sim labs/ooo_compress_bimod
    .out benchmarks/compress95.alpha < benchmarks/compress95.in
```

Extract CPI and IPC:

```
grep -E "sim_CPI|sim_IPC|sim_num_insn|sim_cycle" labs/ooo_*.out
```

Fill Table 2.

Benchmark	Taken CPI	Not-taken CPI	Bimod CPI
`anagram.alpha`			
`compress95.alpha`			

Create a histogram for the two benchmarks and compare the three schemes.

5 Bonus

Write, compile, and run a PISA C program containing a loop that iterates 10 million times. Use the compile-program lab for the compiler setup. Then calculate branch address prediction rate for taken, not-taken, and bimodal predictors.